Seat No._____

GANPAT UNNIVERSITY

M.Sc.IT(AI&ML) Sem-II Regular EXAMINATION
P82A1PYT2 – PYTHON PROGRAMMING-2

Time: 3 Hrs. **April - June 2023** **Total Marks: 60**

1. Figures to the right indicate full marks.
2. Be precise and to the point in your answer

MARKS

Q.1 Answer the following Questions. (Any Five) **30**

1) State different File Object Methods and explain any two with example.
2) Discuss the Python Regular Expression Special Sequences with example.
3) List the different operation performed on Data Frame. Explain isnull(), fillna() and replace() with suitable example.
4) Discuss advantages and disadvantages Data Visualization.
5) Explain the Steps to connect MySQL database in Python using MySQL Connector with programming code.
6) How to access MySQL database using phpmyadmin? Explain in brief.
7) Discuss XML Processing using python module.
8) Write down the python script to show the different widgets of Tkinter using appropriate example.

Q.2 Answer the following Questions. (Any Two) **12**

1) What is Data Frame? Explain the data frame creation from following with suitable examples. (a) CSV file (b) Dictionary
2) Discuss the following graphs with features and explain with figure.
   (a) Bar Graph (b) Histogram
3) Write a python program to fill Student Details form using file read and write operations.

Q.3 Answer the following Questions. (Any Two) **12**

1) What is MySQL? What are the different features of MySQL?
2) Write a python code to perform following operations using appropriate python module. a) Create Database b) Insert Record c) Select Record
3) What is quantifier? Explain the various quantifiers in regular expressions.

Q.4 Answer the following Questions. (Any One) **6**

1) State advantages and disadvantages of GUI Programming.
2) Create python script to email any invitation card using SMTP Protocol.

***** END OF PAPER ******

Seat No._____

GANPAT UNIVERSITY

M.Sc.IT(AI&ML) SEM-II EXAMINATION

P82A2DAT - DATA ANALYTICS USING TOOLS

April - June 2023

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

1. **Answer the following: (ANY SIX OUT OF EIGHT)** **(30)**

A. Write a short note on descriptive analysis.

B. Discuss Data cleaning in brief.

C. Explain moving average and exponential moving average with example in excel.

D. Explain correlation example with EXCEL data analysis tool pack.

E. Describe any three IO nodes in KNIME

F. Explain regression model with example in KNIME.

G. Explain preprocessing example in WEKA.

H. Write a short note on association rule.

2. **Answer the following: (ANY ONE)** **(06)**

A. Explain KDP process in detail.

B. Describe various Data formats.

3. **[A]** **Answer the following: (ANY TWO)** **(08)**

I. Explain all type of ANOVA test in EXCEL.

II. Write a short note on PIE and BAR chart in EXCEL.

III. Explain use of what if analysis in EXCEL with example.

[B] Give full form of ANOVA. **(01)**

4. **Answer the following: (ANY ONE)** **(06)**

A. How many states a node have in KNIME? Explain in brief.

B. Describe any classification example with KNIME in detail

5. **[A]** **Answer the following: (ANY TWO)** **(08)**

I. Explain WEKA environment in detail.

II. Explain any clustering example in WEKA.

III. Explain any five filters in WEKA.

[B] What is default file extension used in WEKA tool? **(01)**

END OF PAPER

Enrollment No.

GANPAT UNIVERSITY

M.Sc.IT (AI&ML) - SEM-II EXAMINATION CBCS (NEW)

P82A3ML1: MACHINE LEARNING - I

April - June 2023

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

1. Answer the following: (ANY SIX FROM EIGHT) **(30)**

a Explain the steps of ML framework for developing model in brief.

b What is unsupervised learning? Explain the classification problem in brief.

c Explain the logistic regression algorithm in detail with proper example.

d What is confusion matrix? Explain in brief.

e Explain polynomial regression in detail with proper example.

f Explain naïve bayes algorithm in brief.

g What is KNN? Explain KNN in detail with proper example.

h Explain any one scaling technique in detail.

2. Answer the following: (ANY ONE) **(06)**

a What is machine learning? Explain supervised machine learning in detail.

b Explain the terms: Overfitting, Underfitting, Bias, Variance

3. Answer the following: (ANY TWO) **(09)**

a What is linear regression? Explain linear regression in detail.

b Explain ROC curve and AUC in detail.

c Explain any two-evaluation parameter of regression in brief.

4. Answer the following: (ANY ONE) **(06)**

a What is regularization? Explain LASSO technique in detail.

b What is regression? Explain multi linear regression in detail.

5. Answer the following: (ANY TWO) **(09)**

a What is validation technique? Explain any two-validation technique in brief.

b Explain support vector machine algorithm in detail.

c Explain decision tree algorithm in detail.

END OF PAPER

Enrollment No._____

GANPAT UNIVERSITY

M.Sc.IT (AI&ML) - SEM-II EXAMINATION Regular Theory

P82B4AI: ARTIFICIAL INTELLIGENCE

Time: 3 Hours] April - June 2023 **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

1.		**Answer the following: (ANY SIX FROM EIGHT)**	**(30)**
	a	How does AI work?	
	b	Rational Agent in AI.	
	c	Explain Informed Search algorithms in detail.	
	d	Explain Alpha – Beta Pruning.	
	e	Explain probability axioms in AI.	
	f	Explain Reasoning with Uncertain Knowledge in AI.	
	g	Write note on Fuzzy logic.	
	h	Write a note on learning.	

2.		**Answer the following: (ANY ONE)**	**(06)**
	a	List out the different type of Agent Explain all with suitable diagram and algorithm.	
	b	List out different four types of problem solving example. And explain all with proper diagram and in form of different properties.	
3.		**Answer the following: (ANY TWO)**	**(09)**
	a	Explain in detail Depth-limited search (DLS).	
	b	Explain A* search.	
	c	Explain Optimal Decisions in Games with proper example.	
4.		**Answer the following: (ANY ONE)**	**(06)**
	a	Explain joint probability distributions in AI.	
	b	Explain First order predicate calculus (FOPL) with example.	
5.		**Answer the following: (ANY TWO)**	**(09)**
	a	Explain the case studies of MYCIN.	
	b	Explain the expert system shells and Vidwan framework.	
	c	Discuss the knowledge acquisition in detail.	

END OF PAPER